

Gonçalo Pereira

Embedded Systems · Network Programming · Backend Development

goncalo.pereira.1021@gmail.com Github Portfolio LinkedIn +351 925 526 272 Lisbon, Portugal

Embedded software engineer working in C on telecom network gateways running embedded Linux. Background in systems and network programming from 42 Lisboa, and hands-on experience designing, shipping and operating a production Go service with its own CI, deployment and monitoring.

EXPERIENCE

- Embedded Software Engineer** · Streambow Dec 2025 – Present · Sintra, Portugal
- Develop embedded C software for telecom network gateway devices running embedded Linux
 - Integrate containerised components with Docker into network diagnostics pipelines
 - Debug networking behaviour through low-level system and packet analysis
- Student Supervisor** · 42 Lisboa Jul 2024 – Oct 2024 · Lisbon, Portugal
- Reviewed and evaluated C and systems programming projects ensuring specification compliance
 - Delivered technical presentations to audiences of 100+ students
 - Coordinated staff workflows supporting student progress initiatives

PROJECTS

- ft_ping & ft_traceroute** · C · Raw Sockets · ICMP · Network Internals 2025
- Reimplemented ping and traceroute using raw sockets and handcrafted ICMP packets
 - Implemented TTL control, checksum computation and round-trip timing without libc abstractions
- malloc** · C · Memory Management · mmap · pthreads 2025
- Thread-safe drop-in replacement for malloc, free, realloc and calloc, built over mmap
 - Pre-allocated TINY/SMALL/LARGE zones, mutex-guarded metadata, allocation-map debug tooling
- Minishell** · C · Process Management · Signals · Parsing 2024
- Built a POSIX-style shell with pipes, redirections, built-ins, environment handling and job signals
 - Designed a tokeniser and AST-based execution pipeline driving fork/exec workflows
- ft_irc** · C++ · TCP Sockets · poll() · Protocol Design 2024
- Built an IRCv3 server handling many concurrent clients from a single poll() loop over non-blocking sockets
 - Implemented channels, operator modes and command parsing; validated against real clients
- dune** · C++ · Systems Architecture · State Machines 2026
- Designing a 45k-line C++ engine for a complex turn-based board game
 - Phase state machine and event bus, with interaction adapters for TTY, scripted and recorded play
- Quest Board** · [dnd.fontao.net](#) · Go · PostgreSQL · React · OpenAPI · Docker 2026
- Designed, shipped and operate a production web app: a React SPA compiled into a single Go binary
 - Contract-first API — one OpenAPI spec generates the Go server interface and the typed TS client
 - OAuth2, local accounts and TOTP 2FA; CI-gated merges, staging and production deploy pipelines
- Homelab Observability** · Prometheus · Grafana · Docker Compose 2026
- Monitoring stack for the above: application, host and per-container metrics, 30-day retention
 - Alert rules provisioned as code; Grafana behind Cloudflare Access, Prometheus never public

EDUCATION

- 42 Lisboa** · Software Development · Systems & Network Programming Oct 2023 – May 2025 · Lisbon, Portugal
- Project-based curriculum covering C/C++, operating systems, networking and algorithms
 - Specialisations in Linux internals and network protocols, and in cybersecurity and reverse engineering

SKILLS

- Languages** C, C++, Go, Python, Rust, JavaScript/TypeScript, x86-64 Assembly, Bash, SQL
- Systems** Embedded Linux, Linux internals, syscalls, signals, process and memory management
- Networking** Raw sockets, TCP/IP, ICMP, protocol implementation, packet analysis (Wireshark, tcpdump)
- Backend** Go (chi, sqlc, pgx), PostgreSQL, REST and OpenAPI, Django, WebSockets, OAuth2/JWT/TOTP
- Infrastructure** Docker, Kubernetes (K3s), Prometheus, Grafana, GitHub Actions CI/CD, NGINX
- Tooling** Git, GDB, Valgrind, Make, contract-driven code generation